Application of topological ideas to the Aharonov–Bohm effect†

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Abstract The Aharonov–Bohm effect is studied to elucidate the existence of global effects in field theory originating from the topology of the field configuration. The problem is formulated in the language of Maxwell's equations, cohomology and principal fibre bundle theory. We offer a motivation for the use of these concepts. In particular, the idealised and realistic descriptions of the Aharonov–Bohm effect are distinguished.

Zusammenfassung Der Aharonov–Bohm Effekt wird studiert in Hinsicht auf die Existenz von globalen Effekten in einer Feldtheorie, die von der Topologie der Feldkonfiguration verursacht werden. Die Feldkonfiguration wird beschrieben mit Hilfe der Maxwell'schen Gleichungen, der Kohomologie Theorie und der Sprache der Hauptfaserbündel. Der Vorteil dieser Beschreibungsweisen für unser Problem wird diskutiert und motiviert. Insbesondere wird die Feldkonfiguration in der idealisierten und der realistischen Beschreibungsweise des Aharonov–Bohm Effektes unterschieden.

1. Introduction

Most effects in physics are restricted to small space–time intervals and any correlation to more distant events fades to insignificance. An apple falling from a tree in Mainz does not influence the detection of W bosons at CERN. However, there are global effects even in everyday physics which depend simply on the topology of the relevant space. For example a theorem in differential topology states that every vector field on the S^2, the two-dimensional sphere, has an even number of zeros. Let the wind on the Earth's surface be a vector field. If we have a calm in Mainz then we know that we have at least one other calm somewhere on the earth.

Topological effects appear in all fields of physics and have especially been noticed in elementary particle physics (Jackiw 1984a, b). The natural language to take into account the global situation of a physical problem is provided by the mathematics of cohomology and principal fibre bundles. To become familiar with these concepts it is advisable to study an elementary example first.

In this article we want to study the Aharonov–Bohm effect (Aharonov and Bohm 1959) and related problems in the framework of these concepts. In §2 we look at the field configuration of the Aharonov–Bohm experiment in the language of Maxwell's equations. In §3 we study the physical consequences of a Young's interference experiment in this set-up and the related effect of flux quantisation in a superconductor. Section 4 closes the introduction with a discussion of Dirac's monopole. In §5 a physical motivation for the use of cohomological arguments and definitions of some concepts are given. In §6 a motivation and definition of the theory of principal fibre bundles is given and in §7 this concept is used to study the monopole. In the last section we formulate our original problem, the field configuration of the Aharonov–Bohm experiment, in the framework of cohomology and principal fibre bundle theory. We focus especially on the question of what the implications of an idealised description of the Aharonov–Bohm effect are. With this pedagogical survey we hope to motivate the use of concepts like cohomology and principal fibre bundle theory to the nonspecialist and experimentalist. Among the many contributions to this problem

† Dedicated to the 50th birthday of our teacher F Scheck.

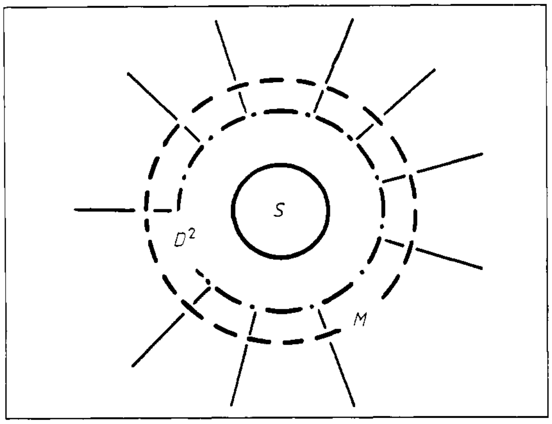


Figure 1. Plane cutting the solenoid. Region D^2 contains the sectional area of the solenoid S, the region M has $B=0$.

we want to mention Yang (1983) and Felsager (1985) which hold a similar point of view.

2. Geometry and field configuration

We are considering a very long solenoid which is positioned along the z axis in a cartesian coordinate frame. Since the endpoints of the solendoid are far away from the centre, the magnetic field outside is arbitrarily small.

We want to study the situation in the x–y plane cutting the solenoid. There we can distinguish an interior region, topologically D^2 (closed disc with radius R) containing the sectional area of the solenoid where the magnetic field $\boldsymbol{B}$ does not vanish, and an overlapping exterior region $M=\mathbb{R}^2\backslash\{$open disc with radius $<R-\varepsilon\}$ where $\boldsymbol{B}$ vanishes (see figure 1). The overlap is $M\cap D^2=S^1\times\varepsilon$. The magnetic flux Φ through D^2 can be connected with the vector potential $\boldsymbol{A}_i$, $\boldsymbol{B}_i=\nabla\times\boldsymbol{A}_i$ via Stokes' theorem

$$\Phi=\int_{D^2}\mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B}_i=\int_{\partial D^2}\mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_i\neq 0$$

where the line integral is evaluated on the boundary of the disc D^2. This line integral can also be considered as a line integral inside M, a region in which $\boldsymbol{B}=0$. Although $\boldsymbol{B}_e=\nabla\times\boldsymbol{A}_e$ vanishes in M, the vector potential $\boldsymbol{A}_e$ in M cannot, because

$$\oint_{\partial D^2}\mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_i=\oint_C\mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_e$$

where C is the boundary ∂D^2 in M.

A solution for $\boldsymbol{A}_e$ is $\boldsymbol{A}_e=(1/er^2)\ (y,\ -x,\ 0)$, where $r=\sqrt{x^2+y^2}$ is the distance from the z axis and e the electric unit charge. We can also determine the vector potential in the interior region $\boldsymbol{A}_i\sim(1/eR^2)(y,\ -x,\ 0)$. R is a constant length. $\boldsymbol{A}_i$ and $\boldsymbol{A}_e$ can be combined into a regular, smooth and z-invariant vector potential $\boldsymbol{A}$ defined on the whole space $\mathbb{R}^3$ with the help of a 'partition of unity':

$$\boldsymbol{A}=f(r)\boldsymbol{A}_i+(1-f(r))\boldsymbol{A}_e.$$

$f(r)$ is a function, with a value of unity inside the solenoid, falling smoothly to zero while still inside D^2 and being zero outside in M.

Now we consider the situation in the exterior region. There we have approximated the true field configuration with zero. $\boldsymbol{A}_e$ differs from the null potential $\boldsymbol{A}_0=0$ only by a local gauge transformation since in both cases $\boldsymbol{B}=0$. Recapitulating possible gauge transformations one immediately finds that $\boldsymbol{A}_{\text{gauge}}=\nabla\lambda$ where λ is an arbitrary, time-independent, two-times differentiable function since $\boldsymbol{B}=\nabla\times\nabla\lambda=0$. But $\boldsymbol{A}_e$ cannot be such a pure gauge, since we would have

$$\Phi=\oint_C\mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_e=\int_C\mathrm{d}\boldsymbol{l}\cdot\nabla\lambda=\lambda(2\pi)-\lambda(0)=0.$$

We have to admit a more general class of gauge transformations. Instead of gauges which are globally the gradient of a function, we admit gauges which are a collection of gradients $\nabla\lambda_i$ of functions λ_i defined piecemeal on a covering $\{U_i\}$ of M. Patching the pieces U_i together gives a smooth vector potential on the whole region, but the functions λ_i cannot be patched together continuously. We call these gauge transformations 'non-trivial'.

The existence of such non-trivial gauge transformations depends on the topology of the space where they are defined, as we will see later. We need non-trivial topology. In our case we can cut M in two parts U_1 and U_2, here described by the polar angle φ

U_1: $\quad \boldsymbol{A}_e=\nabla\lambda_1,$

$$\lambda_1=\frac{\Phi}{2\pi}\tan^{-1}\left(\frac{x}{y}\right),\qquad \varphi\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$$

U_2: $\quad \boldsymbol{A}_e=\nabla\lambda_2,$

$$\lambda_2=\frac{\Phi}{2\pi}\left(\tan^{-1}\left(\frac{x}{y}\right)+\pi\right),\qquad \varphi\in\left[\frac{\pi}{2},\frac{3\pi}{2}\right]$$

and we have

$$\int_C\mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_e=\int_{-\pi/2}^{\pi/2}\mathrm{d}\boldsymbol{l}\cdot\nabla\lambda_1+\int_{\pi/2}^{3\pi/2}\mathrm{d}\boldsymbol{l}\cdot\nabla\lambda_2=\Phi\neq 0.$$

We have indeed found a non-trivial gauge transformation defined inside M. In D^2 the vector potential is not a pure gauge.

3. Physical effects

We want to discuss matter in the background field of the solenoid in the framework of nonrelativistic quantum mechanics.

The principle of local gauge invariance tells us to require invariance of the equation of motion, i.e.

the Schrödinger equation in the background field, under gauge transformations. When we put $A \to A + \nabla\lambda$, the wavefunction acquires an additional phase $\psi \to e^{i\lambda}\psi$. In the case of an ordinary gauge transformation this has no physical implications which it was the intention to achieve of the principle of local gauge invariance. In the case of a non-trivial gauge transformation we can consider the Schrödinger equation, like the gauge transformation, piecemeal. Let us exclude the region D^2 first which we can understand in physical terms as shielding the solenoid against penetration of the wavefunction.

We can study the phase of the wavefunction in the non-trivial gauge configuration when we pass the solenoid on the left-hand side as well as on the right-hand side, as illustrated in figure 2. When the two parts of the wavefunction meet, which they do actually in every point, they superpose to form the physical wavefunction (see figure 2).

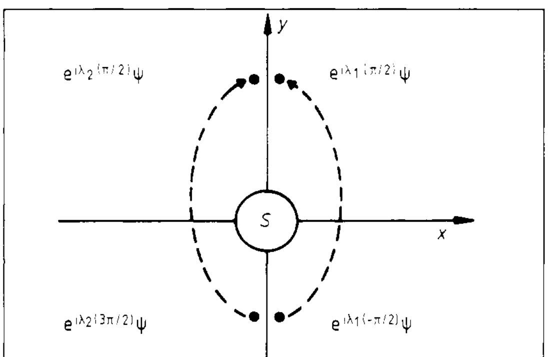
$$\psi = e^{i\lambda_1}\psi'_1 = e^{i\lambda_2}\psi'_2 = e^{i\lambda_1}(\psi'_1 + e^{i\Delta\lambda}\psi'_2)$$

and we have constructive and destructive interference. Looking at a fixed point, e.g. on a screen, the interference situation depends on the phase difference, i.e. the magnetic flux, although the wavefunction only exists in a field-free region. This is the Aharonov–Bohm effect which can be realised experimentally (Bayh 1962). Shielding the solenoid against penetration of matter is not essential since the penetrated and distorted waves would only blur the interference picture. It is only necessary to implement a Young's interference experiment.

The Aharonov–Bohm effect is discussed in relation to the question whether the vector potential itself has a physical reality.

There is a related spectacular effect in the case where we have a superconducting solenoid. We want to consider the wavefunction of the Cooper pairs as the matter field inside the superconductor. The Meissner–Ochsenfeld effect implies that the Cooper pairs must be located outside the magnetic field, which in our geometry is the exterior region M (see figure 3).

Figure 2. Development of the phase of the wavefunction in M.



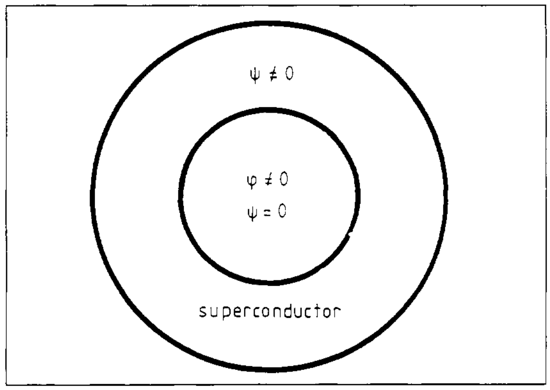


Figure 3. Plane cutting the superconducting solenoid. Regions with non-vanishing flux and non-vanishing wavefunction are specified.

Putting the matter field in a field-free region here is not enforced by geometry, i.e. length of the solenoid, but through the Meissner–Ochsenfeld effect. Taking as an additional input the fact of the axial symmetry of the wavefunction, we see that the phase cannot change arbitrarily when we go around the solenoid because in general we have destructive interference. Since the Cooper pairs should be the sources of our magnetic field, only phase changes of $2\pi n$, $n \in Z$, are admissible. Because of this the magnetic flux is quantised in the superconductor (Deaver and Fairbank 1961). The existence of these two effects confirms the correctness of our description in terms of non-trivial gauge transformations.

4. Real and simulated monopole

To understand the origin of these effects properly we want to move in our system, in the third dimension to one of the ends of the solenoid. In the limit of a very long and very thin solenoid, the endpoint is the source of a radial symmetric magnetic field. It simulates a magnetic monopole.

We are dealing with only a simulation of a magnetic monopole since the flux is not directed outwards in the whole solid angle. The same amount of flux is entering in the vicinity of the z axis such that the total flux through a spherical surface S^2 with the endpoint at its centre is zero. Considering the situation more closely, we can choose an arbitrary, closed, non-intersecting loop C on S^2. C can be considered as a boundary of D_+ as well as of D_- which cover S^2 (see figure 4). The total flux is zero:

$$0 = \int_{S^2} \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B} = \int_{D_+} \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B} + \int_{D_-} \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B}$$

$$= \oint_C \mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_+ + \oint_{-C} \mathrm{d}\boldsymbol{l}\cdot\boldsymbol{A}_- = \oint_C \mathrm{d}\boldsymbol{l}\cdot(\boldsymbol{A}_+ - \boldsymbol{A}_-).$$

The vector potential in D_+ and D_- can differ on C only by a trivial gauge transformation.

Going over to the case of a Dirac monopole we can quote Dirac's (1931) argument of the unobservability, i.e. non-existence, of the field-generating solenoid, which is another expression of the fact that the monopole is in some way in conflict with ordinary electrodynamics. The outwards-directed magnetic flux is not compensated by a solenoid flux and the total flux through S^2 does not equal zero:

$$\int_{S^2} \mathrm{d}\boldsymbol{S}\cdot\boldsymbol{B} = \int_C \mathrm{d}\boldsymbol{l}\cdot(\boldsymbol{A}_+ - \boldsymbol{A}_-) \neq 0.$$

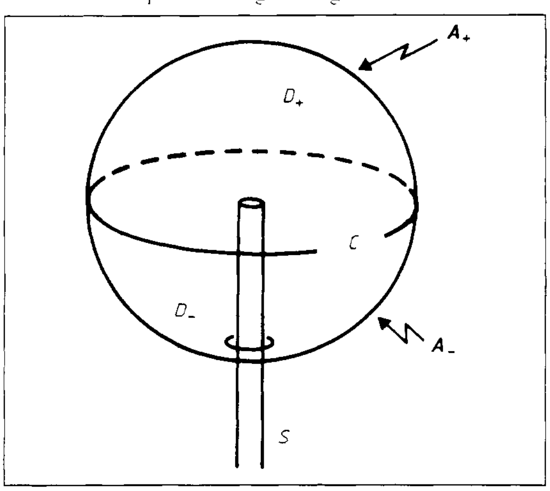
Since the field on C is unique, $\boldsymbol{B}_+|_C = \boldsymbol{B}_-|_C$, the vector potentials $\boldsymbol{A}_+|_C$ and $\boldsymbol{A}_-|_C$ must differ by a non-trivial gauge transformation. Again we cannot have such a vector potential on the whole $\mathbb{R}^3$ as we will see later. We must exclude at least one point: on such a space we have a solution. To be concrete

$$\boldsymbol{A}_\pm = \frac{1}{2r}\frac{1}{z \pm r}\begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} \qquad r = \sqrt{x^2+y^2+z^2}$$

are the piecemeal-defined vector potentials which describe a radial symmetrical field on $\mathbb{R}^3\setminus\{0\}$. The difference $(\boldsymbol{A}_+ - \boldsymbol{A}_-)|_{\mathbb{R}^2\setminus\{0\}}$ looks like the non-trivial gauge transformation familiar from the Aharonov–Bohm effect. But in the former case it was the gauge transformations which had to be defined piecemeal, not the vector potential.

Dirac's requirement that the solenoid is unobservable is a global condition. Locally Maxwell's equations, especially $\nabla\cdot\boldsymbol{B} = 0$, are still valid except for the origin. In this point it is revealed whether $\nabla\cdot\boldsymbol{B} = 0$ is valid or whether we have a magnetic source

Figure 4. Simulation of a monopole-field of configuration at the end of the solenoid. D_+ and D_- cover a sphere around the endpoint cutting all magnetic flux lines.



$\nabla\cdot\boldsymbol{B} = \rho_{\text{mag}}$. Hence we can describe the monopole on almost the whole space in the framework of electrodynamics. We have to introduce a topological defect and describe the phenomenon in the language of electrodynamics on $\mathbb{R}^3\setminus\{0\}$.

We have now seen the phenomenology of some interesting and well known field configurations, but our means of describing them are not yet satisfying. The Aharonov–Bohm effect seems to resemble some aspects of the real monopole although we have a situation completely described by classical electrodynamics.

5. Cohomology and Maxwell's theory

In this part we want to have a look at topology and geometry to profit from the work of the mathematicians. We write Maxwell's equations in the language of differential forms $\mathrm{d}'F = 0$ and $\mathrm{d}'{}^*F = {}^*j$ where F is a differential 2-form over the space–time M^4, ${}^*F$ its dual and d' is Cartan's exterior derivative. In physical terms F contains the antisymmetrical field strength tensor and d' time and space derivatives. In our case where only static magnetic fields exist, the homogeneous equation reduces to $\mathrm{d}B = 0$ where B is the magnetic field 2-form over the space, usually $\mathbb{R}^3$, and the exterior derivative d contains only space derivatives. Then d acts on 0-forms (functions) as a gradient, on 1-forms (vectors) as a curl and 2-forms (pseudovectors) as a divergence. Usually we conclude from the validity of $\mathrm{d}B = 0$ that there is a 1-form A (vector potential) such that $B = \mathrm{d}A$. But we have seen that this conclusion is not correct if the topology of the space is nontrivial, that means e.g. $\mathbb{R}^3\setminus\{0\}$. $\mathrm{d}B = 0$ means that B is a *closed* form (here B is a form of arbitrary degree). If there is a form A such that $B = \mathrm{d}A$, then B is an *exact* form. Every exact form is closed since $\mathrm{d}\cdot\mathrm{d} = 0$, but not vice versa. In mathematical terms the question 'is there a potential for B?' means 'is B an exact form?'

To handle this problem there is the notion of the cohomology group (Bott and Tu 1982, Nash and Sen 1983)

$$H^q(M) := \text{Ker d}/\text{Im d}.$$

This needs some explanation:

Ker d are the closed q-forms $\{\lambda \in \Omega^q(M) | \mathrm{d}\alpha = 0\}$

Im d are the exact q-forms

$$\{\gamma = \mathrm{d}\beta \in \Omega^q(M) | \beta \in \Omega^{q-1}(M)\}.$$

The quotient indicates the equivalence relation

$$\alpha + \mathrm{d}\beta \sim \alpha$$

i.e. all closed q-forms which differ by an exact form are identified and $\alpha + \mathrm{d}\beta$ belongs to the same equivalence class $[\alpha]$. Closed forms, which differ by a non-exact form, belong to different equivalence classes. If there are non-exact closed forms of degree q, then $H^q(M)\neq[0]$. The $H^q(M)$ are determined completely by the topology of M. Non-trivial topology of M means non-trivial $H^q(M)$ for some q. We can discuss the physical implications in two examples.

5.1. $M=\mathbb{R}^3$

Mathematics provides that $H^n(\mathbb{R}^k)=0$, $n>0$, $k>0$. $B\in[0]=H^2(\mathbb{R}^3)$ is closed, i.e. $dB=0$, and differs only by an exact form dA from zero, i.e. $B=dA$. For every field configuration there is a vector potential.

5.2. $M=\mathbb{R}^3\backslash\{0\}$

Since $\mathbb{R}^3\backslash\{0\}\simeq\mathbb{R}\times S^2$ homotopic to S^2 we have now $H^2(\mathbb{R}^3\backslash\{0\})\cong H^2(S^2)\neq[0]$ and there is not always a vector potential for B.

In the case where B is a trivial element we have with the help of Stoke's theroem

$$B\in[0]\Leftrightarrow\int_K B=\int_K dA=\int_{\partial K=0} A=0$$

for all closed 2-dimensional surfaces $K\subset M$. There is no flux. However, considering the field of a magnetic monopole

$$B_m=\frac{1}{4\pi r^3}(x_1\,dx_2\times dx_3-x_2\,dx_1\times dx_3+x_3\,dx_1\times dx_2)$$

$$r=|\boldsymbol{x}|$$

which belongs to a vector $\boldsymbol{B}=\boldsymbol{e}_r/4\pi r^2$. We can calculate easily, with K a sphere around the origin,

$$\int_K B_m=1.$$

Therefore $[0]\neq[B_m]\in H^2\mathbb{R}^3\ \{0\})$ and there is no global vector potential for the monopole.

It is still possible to keep the notion of the vector potential in general cases like §5.2 at the expense of introducing a new structure in our theory. Since the Aharonov–Bohm effect indicates that the vector potential has physical relevance, this seems reasonable.

6. Principal bundles and Maxwell's theory

This new structure provides the definition of exact forms Ω on a new space P which can be related to closed forms F and M which are not necessarily exact. The main idea is to enlarge the space M to a space E to circumvent topological obstructions. In this chapter we use the more general field strength 2-form F instead of B.

To introduce the notion of the principal fibre bundle we consider simple fibre bundles first. A carpet is an illustrative example of a simple fibre

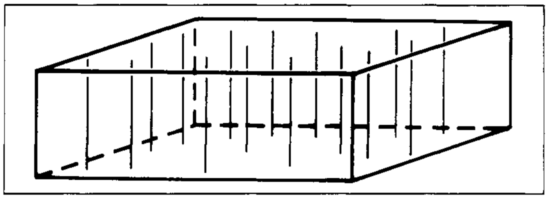


Figure 5. A carpet illustrating base space and fibres of a trivial fibre bundle.

bundle. In mathematical language a carpet consists of a base space M, the total space P and a mapping π: $P\rightarrow M$, which maps every point in P to a point of the base space. All points $\pi^{-1}(b)\in P$ which belong to a point $B\in M$ form the fibre Y in b. Any fibre bundle is locally trivial, i.e. should locally look like the product space $M\times Y$ (see figure 5). In our example we even have a globally trivial situation, i.e.

$$\text{carpet}=\text{rectangle}\times\text{fibre}.$$

Fibre bundles become interesting when they are not globally trivial. A nice example is the comparison between a cylindrical strip and the Möbius strip. Here the base space and the fibre are S^1 and an interval (see figure 6). Sitting in the neighbourhood of a point a or b both fibre bundles seem to be equivalent; we see rectangles in figure 6. Only a global view detects that one strip is glued together simply while the other is twisted. Along these lines we can in general describe a fibre bundle by specifying patches of the base space over which the bundle is trivial and prescriptions for how the fibres are glued together where the patches overlap.

Up to now a typical fibre was a line or more generally, a vector space. If we use a Lie group G as a typical fibre (or actually its group manifold which is usually no vector space anymore) we are close to the concept of a principal fibre bundle. A principal fibre bundle is a fibre bundle with total space P, base space M, Lie group G as typical fibre (structure group) and as additional structure a right action Φ of G on P

$$\Phi:P\times G\rightarrow P$$

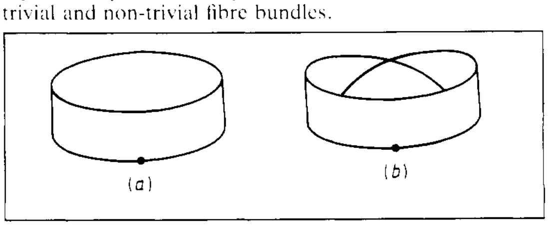
which is 'free', i.e. if

$$u\cdot g=u \qquad \text{for} \qquad u\in P, \qquad g\in G\Rightarrow g=e$$

and acts only on the fibre, i.e.

$$\pi(u\cdot g)=\pi(u).$$

Figure 6. Cylindrical strip (a) and Möbius strip (b) as trivial and non-trivial fibre bundles.



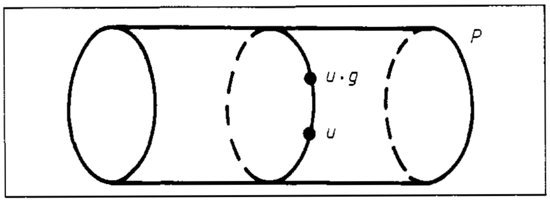


Figure 7. Total space of the principal fibre bundle $\mathbb{R} \times U(1)$ and group action on it.

A simple example is the trivial principal fibre bundle $\mathbb{R} \times U(1) \simeq \mathbb{R} \times S^1$, the total space of which is the infinite cylinder and whose group action is a turning of the cylinder. Note the free action: no point stays fixed except for $g = e$ (see figure 7). Also for the principal fibre bundle there is a description in terms of patches and glueing prescriptions. An example of a non-trivial principal fibre bundle is not easily given since our imagination is readily exhausted. On such a principal fibre bundle there exists a 1-form ω with $\Omega = \mathrm{d}\omega$, hence $\Omega \in [0] \in H^2(P)$; ω corresponds to the vector potential A on M and Ω to the field strength F on M. To perform this projection we have to define a local section.

These are mappings σ_i: $U_i \subset M \rightarrow P$ with $\pi \cdot \sigma_i = id$. The $\{U_i\}$ cover M. Only in the case of a trivial principal fibre bundle do we have a global section. In our example of $\mathbb{R} \times S^1$ we can have the picture of figure 8. The sections σ and $\tilde{\sigma}$ differ by a local group transformation. In the case of local sections (U_1, σ_1), (U_2, σ_2) we have the picture of figure 9. On the overlap $U_1 \cap U_2$ the sections differ by a group transformation. A choice of a section is also called a trivialisation of the principal fibre bundle. For a trivial bundle there exists a global trivialisation (global section). A non-trivial bundle has only local trivailisation (local sections). With a local section σ_i we can induce a mapping of forms

$$\sigma_i^*: \Lambda^q(P) \rightarrow \Lambda^q(U_i)$$

and our field strength F on $U_i \subset M$ is defined by

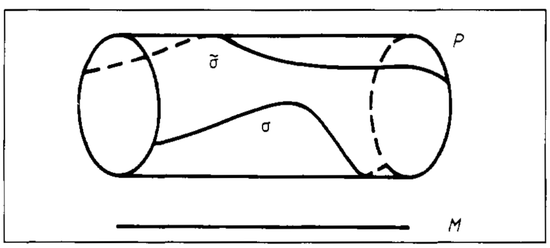
$$F|_{U_i} := \sigma_i^* \Omega$$

and the vector potential by

$$A|_{U_i} := \sigma_i^* \omega.$$

Considering local field strengths and vector potentials on U_1 and U_2 they differ on $U_1 \cap U_2$ by an

Figure 8. Global sections (M, σ) and $(M, \tilde{\sigma})$ in P differing by local group transformations.



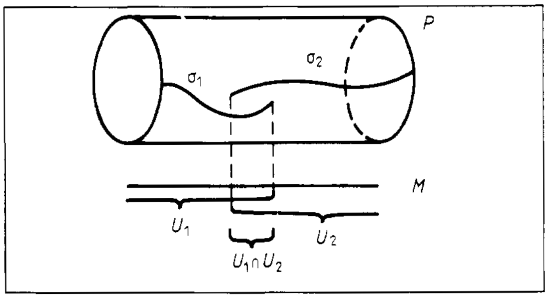


Figure 9. Local sections (U_1, σ_1), (U_2, σ_2) in P differing by a group transformation on the overlap.

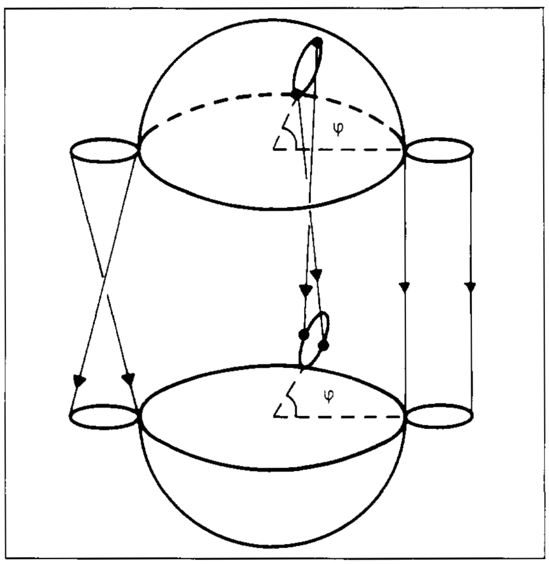
induced group transformation, a gauge transformation. But F is gauge covariant; that is F can still be defined globally. However, A is not gauge covariant, it has to be defined locally! Only in the case of a trivial principal fibre bundle do we have a globally defined vector potential on M.

The advantage of the concept of the principal fibre bundle is to have instead of many local equations $F|_{U_i} = \mathrm{d}A|_{U_i}$ only one global equation $\Omega = \mathrm{d}\omega$ which determines the local quantities F and A with the help of the global structure of the principal fibre bundle.

Ω and ω are actually pure geometrical quantities known as (fibre-) curvature and connection. The principal fibre bundle theory connects geometry with gauge theory.

We finally want to add some remarks on how one characterises the global structure of different principal fibre bundles. Here the fibre curvature Ω plays an important role. With the help of a skilful mapping, Ω becomes a set of polynomials of our field

Figure 10. Prescription to glue the S^1 fibres (small circles) along the overlap of D_+ and D_- to obtain the Hopf bundle.



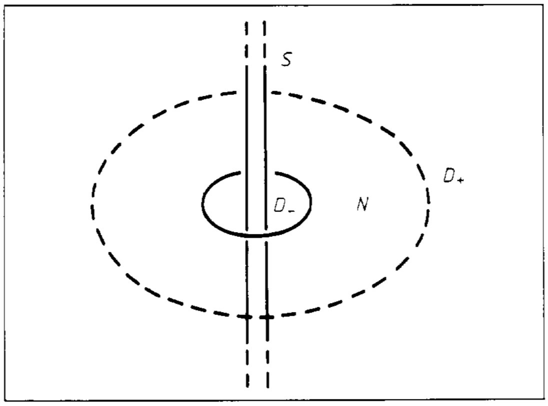


Figure 11. View of the plane cutting the solenoid S. The plane is divided in the region D_- containing the solenoid flux, the field-free region N and the region D_+ containing the returning flux.

strength F on M. These forms, the Chern forms, are integrated over M, and we get the integer-valued Chern numbers c_i, which are topological invariants of the principal fibre bundle. The simplest of them are the first and second Chern numbers:

$$c_1 = \frac{1}{2\pi} \int_{M^2} \operatorname{tr} F,$$

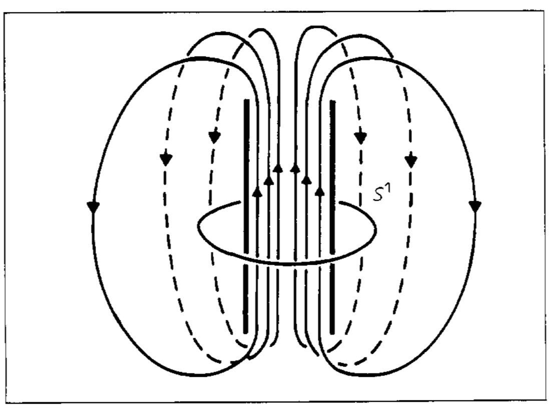
$$c_2 = \frac{1}{8\pi^2} \int_{M^4} \operatorname{tr}(F \times F)$$

which are respectively defined for two-dimensional and four-dimensional compact base spaces without boundary.

7. The monopole in the language of principal fibre bundles

Now we can describe the monopole in the language of the principal fibre bundles. To satisfy the free

Figure 12. Topology of the field configuration of a finite, realistic solenoid. The field-free region is the S^1 of the zeros of the vector field.



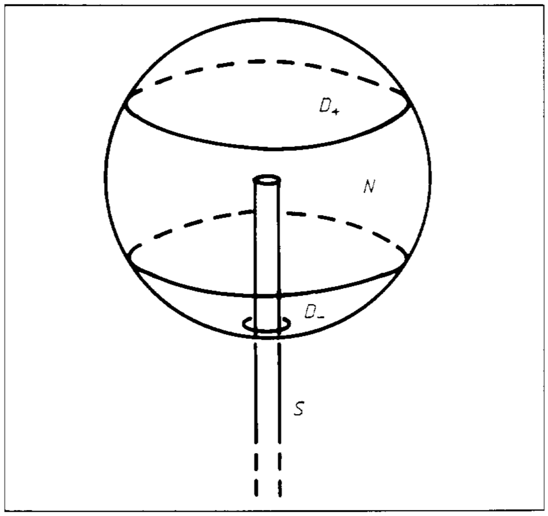


Figure 13. Closed surface cutting the solenoid. Situation of figure 11 with the point at infinity mapped to the north pole of the sphere.

Maxwell equation $dB=0$ everywhere, we have to exclude the origin. Our base space becomes $\mathbb{R}^3\setminus\{0\} \simeq S^2 \times \mathbb{R}^+$. The global structure of our principal fibre bundle is invariant under deformations. Therefore we can restrict ourselves to S^2 as the base space. Since we have not found a global vector potential A for the monopole, we have to construct a non-trivial principal fibre bundle over S^2 with the structure group $U(1) \simeq S^1$. We try to construct the bundle with patches and glueing prescriptions.

As we mentioned before, we have to cover our base space with patches on which the bundle is trivial and then glue the fibres together appropriately. We cut S^2 into two discs D_+ and D_-. It is a fact that any bundle over a disc must be trivial. If we glue the fibres on the boundary of the discs properly, we get the monopole bundle.

Going around the boundary we start glueing the fibres as they are. Proceeding with an angle φ on the boundary, we glue the S^1 fibres turned by an angle φ. Completing the way around the boundary, the fibres are glued together turned by an angle of 2π. Hence the procedure is continuous (see figure 10). We have obtained the Hopf bundle with a total space of S^3. Other possibilities come from turning and glueing the fibres an integral number of times 2π while going around the boundary.

A characterisation of these prescriptions is given by the first Chern numbers $c_1 = (1/2\pi)\int_{S^2} F$ which count the number of unit monopole charges in the region.

8. Aharonov–Bohm effect

Now we want to understand the Aharonov–Bohm effect in cohomological terms and in terms of principal fibre bundle theory.

Before we turn to the real situation we want to study an idealised description. We definitely want to have $\mathrm{d}B=0$ to be valid in the whole space $\mathbb{R}^3$, especially in $\mathbb{R}^2$. Our situation will be described by a bundle which is defined over a space without holes. Like the bundles over the discs on our way to construct the monopole bundle, this bundle is necessarily a trivial bundle.

We also want to have a region in which $B=0$ is valid. The idealisation concerns the range of this region. In §2 it was $\mathbb{R}^3\backslash\mathbb{R}=(\mathbb{R}^2\backslash\{0\})\times\mathbb{R}$. The restriction to the two-dimensional case does not change the situation since

$$H^1(\mathbb{R}^3\backslash\mathbb{R})\simeq H^1((\mathbb{R}^2\backslash\{0\})\times\mathbb{R})\simeq H^1(\mathbb{R}^2\backslash\{0\})\neq[0].$$

Also the restriction to the case of a finite region with $B=0$ does not change the cohomological situation since

$$H^1(\mathbb{R}^2\backslash\{0\})\simeq H^1(S^1\times I)\simeq H^1(S^1)\neq[0]$$

with I a finite interval (thickness of the region with $B=0$). However, it is better to consider the case of a finite field-free region because we need to take account of the magnetic flux coming back and crossing the x–y plane far away from the origin to have the Maxwell equation globally valid. We cut the space $\mathbb{R}^2$ in three regions (see figure 11): the region D_- containing the solenoidal flux, the field-free region N and the region D_+ containing the returning flux and extending to infinity. We now have a trivial bundle $P\to\mathbb{R}^2$ which decomposes into bundles $P=P_1\amalg P_2\amalg P_3$ ($\amalg$ = disjoint union)

$$\begin{array}{ccccccc} P & = & P_1 & \amalg\ P_2 & \amalg & P_3 \\ \downarrow & & \downarrow & \downarrow & & \downarrow \\ \mathbb{R}^2 & = & D_- & \amalg\ N & \amalg & D_+ . \end{array}$$

The mappings are the bundle projections of §6. We can define local trivialisations of P. Our choice of A_i and A_o (here considered as 1-forms on D_- and N) in §2 corresponds to local trivialisations in D_- and N of a 1-form ω defined on P. These local trivialisations are connected by a local gauge transformation. In §2 it was a non-trivial gauge transformation. To complete the trivialisation of the bundle P we also have to define a local trivialisation in D_+ appropriately.

Since the bundle P is trivial, there exists a global trivialisation. We had started to construct one in §2 with our definition of a vector potential defined by smoothing and adding A_i and A_o with the help of a partition of unity. In the same manner we can attach the vector potential in D_+. In the case of non-trivial bundles like a magnetic monopole this cannot be done.

We focus on the local trivialisation of $P_2\to N$. The cohomology group $H^1(N)$ of this field-free region is not trivial. Hence there exist closed, non-exact 1-forms. The local trivialisation of $P_2\to N$ corresponds to a vector potential which belongs to a specific element of the cohomology group $H^1(N)$. This element is determined by the fact that the bundle $P_2\to N$ is a sub-bundle of the bundle $P\to\mathbb{R}^2$. Calculating line integrals we see that we get non-trivial elements of $H^1(N)$ if there is a magentic flux through the solenoid described by the vector potential in $P_1\to D_-$.

Note that in §2 we also needed the validity of Maxwell's equations in the space $\mathbb{R}^2$, expressed by the line integrals, to necessitate the non-trivial transformations corresponding to the non-exact 1-forms. Here, we immediately see that our non-trivial gauge transformations are classified by the elements of the cohomology group $H^1(N)$ which are parametrised by real numbers or in physical terms by the magnetic flux through the solenoid.

A bundle like $P_2\to N$ with vanishing field is called a 'flat bundle'. Although the sub-bundle $P_2\to N$ is flat, it still contains the information about the flux through the solenoid. This is the consequence of the global validity of Maxwell's theory expressed in the language of principal fibre bundles.

If we do not start from the global situtation $P\to\mathbb{R}^2$ but consider this flat, trivial bundle $P_2\to N$ itself, we perhaps do not see the non-trivial effect of a phase shift of the wavefunction at first sight. But another mathematical observation points in the right direction. Flat bundles can still have non-trivial discrete holonomy groups, if and only if they are defined over topologically non-trivial spaces. The holonomy group, in general, describes the change of the direction of a vector after a parallel transport on the base space around a closed loop. Hence the holonomy group must be a subgroup of the structure group. In our case the 'vector' is the phase of the wavefunction and the transport goes around the hole D_-. The possible occurring holonomy group are $\mathbb{Z}$ and $\mathbb{Z}_n$.

Studying the realistic case we first look for a region with a vanishing field. There is still a S^1 region where precisely $B=0$: the zero line of the circulating magnetic field (see figure 12). The cohomology of S^1 is the same as the cohomology of $(\mathbb{R}^2\backslash\{0\})\times\mathbb{R}$. Hence the existence of a non-trivial gauge transformation in the field-free region accounting for the phase shift of the wavefunction is guaranteed. We have a degenerate case compared with the idealised situation. It should be mentioned that there is still a topological difference between the spaces with vanishing field in the idealised and the realistic case. Cohomology is a classification, but not a sufficient classification for the topology of a space. The physical consequences which are related to this difference exceed the scope of this paper.

Since usually this is a finite region $S^2 \times \varepsilon$, ε small interval, where the field is arbitrarily small, the effective theory indicated in figure 13 describes our situation satisfactorily. We have a finite solenoid and a finite region N over which we have a flat bundle. It is the situation of figure 11 with the points at infinity mapped into the north pole of a sphere.

Even in the case of a superconducting coil the statement of $B = 0$ inside the superconducting material is an approximation and strictly true only on a S^1 since the field strength falls off exponentially going inside, starting from the surface.

Finally, we want to remark upon what we call a topological effect in our theory without topological defect. The geometry of the solenoid enforces the existence of a flat bundle over a topologically nontrivial subspace. This is the common point in the realistic and in the idealised (effective) description. Here the vector potentials acquire physical reality in expressing locally the global physical situation.

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